GRE Quantitative Reasoning

모의고사 Set 11 — 문제지

최신 문제(뒷번호) · S2 고난도

구성
Section 1: 12문항 / 21분
Section 2: 15문항 / 26분
합계 27문항 / 47분 · 계산기 사용 가능

대소비교(Quantitative Comparison) 선택지 — 모든 QC 문제 공통
(A) Quantity A is greater. (B) Quantity B is greater.
(C) The two quantities are equal. (D) The relationship cannot be determined from the information given.

안내
- Indicate all (복수선택) 문제는 해당하는 것을 모두 고르세요. 모두 맞아야 정답입니다.
- 주관식 분수 답은 약분하지 않아도 됩니다.
- 문제 이미지 속 번호는 문제집 원번호입니다. 문항 번호는 보라색 'Question N'을 기준으로 합니다.
- 정답과 해설은 별도 파일(Set 11 해설지)에 있습니다.

답안 기록표

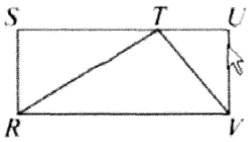
S1	1	2	3	4	5	6	7	8	9	10	11	12
답												

S2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
답															

Section 1 12 Questions · 21 Minutes

Question 1 대소비교

828.



RSUV is a rectangle.
Quantity A: The length of ST / The length of TU
Quantity B: The area of △RST / The area of △TUV

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 2 대소비교

966. $-4<x<2$ and $-2<x<4$
Quantity A: $|x|$
Quantity B: 2

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 3 대소비교

588. n is a positive integer.
Quantity A: The remainder when $n(n^2-1)$ is divided by 6
Quantity B: 1

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 4 대소비교

1109. Quantity A: The ones digit of 208^{208}
Quantity B: 7

(A) Quantity A is greater.

(B) Quantity B is greater.

(C) The two quantities are equal.

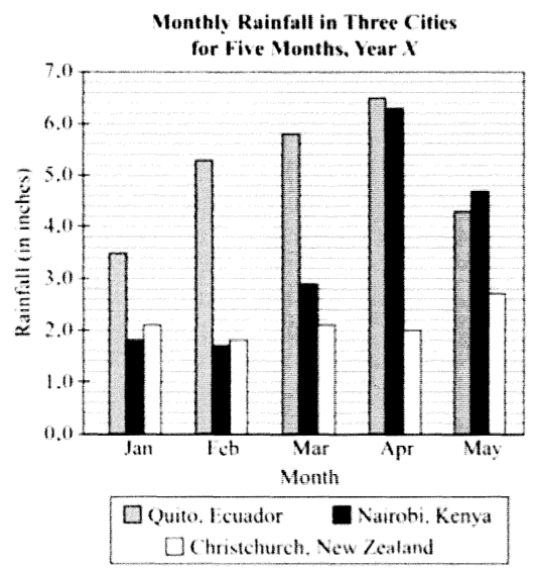
(D) The relationship cannot be determined from the information given.

Question 5 객관식

406. The volume of a red container is 3 times the volume of a green container. What fraction of the green container is equal to 1/12 of the volume of the red container?
A1/36
B1/12
C1/4
D1/3
E1/2

(A) (B) (C) (D) (E)

Questions 6 to 8 are based on the following data.



Question 6 객관식

780. The total annual rainfall during year X in Nairobi was 29.9 inches. Approximately what percent of the total annual rainfall occurred during April?
A45%
B40%
C30%
D25%
E20%

(A) (B) (C) (D) (E)

Question 7 객관식

781. For the five months, the median of the values shown for monthly rainfall, in inches, for Quito was which of the following?
A4.3
B4.8
C5.1
D5.3
E5.8

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Question 8 객관식

782. In Christchurch, the ratio of the number of inches of rainfall during May to the number of inches of rainfall during February was closest to which of the following?
A2.3 to 1
B1.8 to 1
C1.5 to 1
D1.3 to 1
E0.7 to 1

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Question 9 객관식

1125. The degree of a polynomial in x is the greatest exponent of x when the polynomial is written as a sum of terms such as a_ix^i, where i is a positive integer and a_i, is a number. For example, the degree of $9x^3+3x^5-100x$ is 5. What is the degree of the polynomial $(x+4x^3)^2(x^5-7)$?
A10
B11
C12
D13
E14

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Question 10 주관식

749. A stone was dropped into a still pond and produced concentric circular ripples on the surface of the water. The radius of the outermost ripple increased at a constant rate of x feet per second. If the area of the circular region enclosed by the outermost ripple was 400π square feet 10 seconds after the stone hit the water, what is the value of x?

Question 11 복수선택

1278. The perimeter of rectangle ABCD is 20, and the length of each side is an integer. Which of the following values could be the length of diagonal AC?
Indicate all such values.
A. $\sqrt{82}$
B. $\sqrt{68}$
C. $\sqrt{58}$
D. $\sqrt{52}$

Indicate all that apply. [A] [B] [C] [D]

Question 12 객관식

1023. In a government's current budget, 66 percent of the budget is allocated to the payment of entitlements and interest on the government's debt. If the dollar amount of the allocation for entitlements and interest on the government's debt were to remain the same, but all the remaining allocations in the budget were to be reduced by 10 percent of their dollar amounts, by what percent would the entire government budget be reduced?

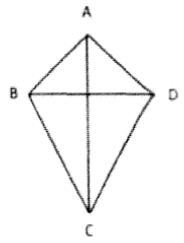
A1.5%
B3.4%
C5.0%
D6.0%
E6.6%

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Section 2 15 Questions · 26 Minutes

Question 1 대소비교

845.



Quantity A: The perimeter of ABCD
Quantity B: The sum of the lengths of diagonals AC and BD

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 2 대소비교

840. $x>y>\sqrt{2}$
Quantity A: xy
Quantity B: x+y

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 3 대소비교

592. a, b, c, d, e, f, g, h, i, j
Let S be the set of all 10-letter sequences consisting of the 10 letters above.
Let T be the set of all 8-letter sequences consisting of the first 8 letters above.
Quantity A: The number of sequences in S in which f, b, h, and g appear consecutively in that order
Quantity B: The number of sequences in T in which f and b appear consecutively in that order

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 4 대소비교

748. In a list of 15 numbers, 8 numbers are greater than 50.4, and 8 numbers are less than 54.6.
Quantity A: The median of the list of numbers
Quantity B: 52.5

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

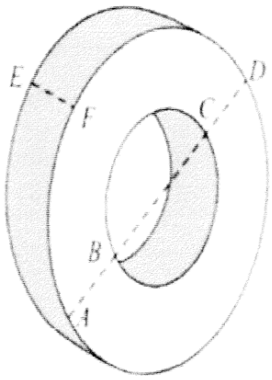
Question 5 대소비교

582. Consider all the permutations of the letters in the word MISSISSIPPI, where I is a vowel and the other letters are consonants.
Quantity A: The number of permutations in which all the vowels are next to each other
Quantity B: The number of permutations in which all the consonants are next to each other

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 6 객관식

1121.



The figure shows a wooden toy wheel made by removing a smaller right circular cylinder from a larger right circular cylinder so that the cylinders are centered around the same axis. Line segments BC and AD are diameters of the smaller and larger cylinders, respectively, and line segment EF represents the width of both cylinders. The lengths, in centimeters, of BC, AD, and EF are 3, 7, and 2, respectively. Approximately what is the volume, in cubic centimeters, of the wood contained in the finished wheel?
A14
B20
C31
D63
E77

(A) (B) (C) (D) (E)

Question 7 객관식

587. All of the codes used to unlock a certain type of jewelry box are different arrangements of the letters iin the name "MARSHA," where both the first and last letters in each code are "A"s. How many such 6-letter codes are possible?
A24
B16
C12
D8
E6

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Question 8 객관식

398. Pat has five matched pairs of socks and no two of the pairs are the same color. If Pat selects two socks simultaneously and at random, what is the probability that the two socks selected will be a matched pair?
A1/2
B1/4
C1/5
D1/9
E1/10

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Question 9 복수선택

496. Of the 120 first-semester students entering a certain college, 3/5 registered for a statistics class and 1/2 registered for a calculus class. Which of the following could be the number of first-semester students entering the college who registered for a statistics class but did not register for a calculus class?
Indicate all such possible numbers.
A0
B6
C12
D30
E60
F72
G90

Indicate all that apply. [A] [B] [C] [D] [E] [F] [G]

Question 10 주관식

1321.



In the figure, if n−t=80, what is the value of n?

Question 11 객관식

363. If x=2y+1, and y=2w, where w, x, and y are integers, which of the following must be an odd integer?
Axy+w
Bxy+w+1
C(x+y)w
Dwy+x
Ewx+y

A B C D E

Question 12 객관식

835. If n and m are positive integers and m is a factor of 2^6, what is the greatest possible number of integers that can be equal to both 3n and $2^6/m$?
AZero
BOne
CThree
DFour
ESix

A B C D E

Question 13 복수선택

711. For each value x in a list of values with mean m, the absolute deviation of x from the mean is defined as |x-m|.
A certain online course is offered once a month at a university. The number of people who register for the course each month is at least 5 and at most 30. For the past 6 months, the mean number of people who registered for the course per month was 20. For the numbers of people who registered for the course monthly for the past 6 months, which of the following values could be the sum of the absolute deviations from the mean?
Indicate all such values.
A100
B90
C60
D30
E10

Indicate all that apply. A B C D E

Question 14 객관식

692. Set R={x, y, z, 2.7, 3.8, 5.5}
Set S={x, y, z, -2.7, -3.8, -5.5}
If the average (arithmetic mean) of the 6 numbers in list R is r and the average of the 6 numbers in list S is s, then r-s is the average of which of the following lists?
Ax, y, z
B2.7, 3.8, 5.5
C-2.7, -3.8, -5.5
D–x, -y, -z
E-2.7, -3.8, -5.5, 2.7, 3.8, 5.5

A B C D E

Question 15 객관식

726. In a group of 10 people, the median height is 70 inches, the average (arithmetic mean) height is 70.5 inches, and the range of the heights is 12 inches. If an additional person who is 74 inches tall joins the group, which of the three statistics must change?
AAverage only
BMedian only
CRange only
DAverage and median
EAverage and range

A B C D E